

MANIPAL UNIVERSITY JAIPUR
Department of Data Science and Engineering
Course Hand-out

A. Basic Details:

Programme Name:	B.Tech. (DS)
Course Name:	Social Network Analysis
Course Code:	DS4145
LTPC (<i>Lecture Tutorial Practical Credits</i>):	[3 0 0 3]
Session:	July – November 2025
Class:	VII Semester
Course Coordinator:	Dr. Aprna Tripathi
Course Instructor(s):	Dr. Aprna Tripathi
Additional Practitioner(s) – if any (<i>Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.</i>):	Nil

B. Introduction: Analysing the Social Web provides a framework for the analysis of public data currently available and being generated by social networks and social media, like Facebook, Twitter, and Foursquare.

C. Course Outcomes:

<i>CO Statement</i>	<i>CO</i>	<i>Level</i>	<i>Target Attainment %</i>	<i>Target Attainment level</i>
Presents interactive social applications on the web, and the types of analysis that are currently conducted in the study of social media,	DS4145.1	3	80%	3
Analyse the basics of network structures for beginners, including measuring methods for describing nodes, edges, and parts of the network,	DS4245.2	3	80%	3

Evaluate the major categories of social media applications or phenomena and shows how the techniques presented can be applied to analyse and understand the underlying data	DS4145.3	3	80%	3
Understand information visualization, particularly network visualization techniques, and methods for using them to identify interesting features in a network, generate hypotheses for analysis, and recognize patterns of behaviour	DS4145.4	2	75%	2

Information about attainment levels:

Attainment (%)	Level
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes:

[PO.1]. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

[PO.2]. Problem analysis: Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

[PO.3]. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

[PO.4]. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5]. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

[PO.6]. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

[PO.7]. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8]. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

[PO.9]. Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

[PO.10]. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11]. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12]. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

[PSO.1]. Understand the role of mathematics, statistics, and AI techniques in the field of data science & engineering.

[PSO.2]. Apply the acquired knowledge and expertise to perform data analytics tasks for multidimensional datasets.

[PSO.3]. Develop effective and scalable industrial solutions for real-world socio-economic problems using data analytics tools and techniques.

PROGRAM EDUCATIONAL OBJECTIVES

[PEO.1]. Acquire and improve data science abilities to compete in a potential job as a data scientist and analyst.

[PEO.2]. Establish entrepreneurial skills to provide new and scalable solutions for tackling real-life complex data processing methodologies, paving the way for startups and self-employment.

[PEO.3]. Pursue higher studies to carry out research and development in the fields of advanced data science and artificial intelligence.

[PEO.4]. Integrate sensitivity towards all facets of ethical, environmental, and social concerns with holistic development.

E. Assessment Plan:

Criteria	Description	Maximum Marks
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Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Two quizzes (both compulsory) and two assignments/ one research paper (both mandatory to submit on time).	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100
Attendance (Formative)	A minimum of 75% attendance is required for a student to be eligible to appear in the End Semester Examination. The permissible 25% absence covers all types of leaves, including medical leave.	

F. Syllabus:

Introduction to Social Web, Nodes, Edges and Network Measures, Describing Nodes and Edges, Describing Networks, Layouts, Visualizing network features, The role of Tie strength, Measuring Tie strength and its network structures, network propagation, Link prediction, entity resolution, Case study, Introduction to community discovery, communities in context, quality functions, The Kernighan-Lin algorithm, Agglomerative algorithms, spectral algorithms, multi-level graph partitioning, Markov clustering, Other approaches, Introduction to social influence, Influence related statistics, social similarity and influence, Homophily, Existential Test for social

influence, Influence and actions, Influence and interactions, influence maximization in viral marketing.

Textbooks:

1. Jennifer Golbeck., Analysing the Social Web, Morgan Kaufmann publications, 2013
2. Charu C. Aggarwal, Social Network Data Analytics, Springer publications, 2011

References:

1. John Scott, Social Network Analysis, (3e), Sage publications limited, 2013
2. Jay Goldman, Facebook Cookbook, O'Reilly, 2009
3. Shamanth Kumar, Fred Morstatter, Huan Liu, Twitter Data Analytics, Springer publications, 2013

G. Lecture Plan:

Lecture No.	Topic	Session Outcome	Corresponding CO	Mode of Delivery	Mode of Assessing CO
1	Introduction to Social Web	Understand the concept of the Social Web.	DS 4145.1	Lecture-based teaching-learning	Assignment, Quiz
2	Nodes, Edges and Network Measures	Define nodes and edges in a network.	DS 4145.1	Lecture-based teaching-learning	Assignment, Quiz
3	Describing Nodes and Edges	Calculate basic network measures.	DS 4145.1	Lecture-based teaching-learning	Assignment, Quiz
4	Describing Nodes and Edges2	Calculate basic network measures.	DS 4145.1	Lecture-based teaching-learning	Assignment, Quiz
5	Describing Networks, Layouts	Analyze different network topologies	DS 4145.1	Lecture-based teaching-learning	Assignment, Quiz
6	Describing Networks, Layouts2	Analyze different network topologies	DS 4145.1	Lecture-based teaching-learning	Assignment, Quiz
7	Visualizing network features,	Apply appropriate layouts for network visualization.		Lecture-based teaching-learning	Assignment, Quiz
8	The role of Tie strength,	Understand the concept of tie strength.	DS 4145.1	Lecture-based teaching-learning	Assignment, Quiz
9	Measuring Tie strength and its network structures,	Analyze the impact of strong and weak ties on network structure.	DS 4145.1	Technology-based learning	Quiz, Assignment
10	Measuring Tie strength and its network structures2	Analyze the impact of strong and weak ties on network structure.	DS 4145.1	Technology-based learning	Quiz, Assignment
11	network propagation,	Describe different network propagation models.	DS 4145.1	Lecture-based teaching-learning	Assignment, Quiz
12	Link prediction	Apply link prediction techniques to identify potential connections.	DS 4145.1	Technology-based learning	Assignment, Quiz
13	Link prediction2	Apply link prediction techniques to identify potential connections.	DS 4145.1	Technology-based learning	Assignment, Quiz

14	entity resolution	Apply appropriate layouts for network visualization.	DS 4145.2	Technology-based learning	Assignment, Quiz
15	Introduction to community discovery,	Define community discovery in networks.	DS 4145.2	Technology-based learning	Assignment, Quiz
16	communities in context	Understand the concept of tie strength.	DS 4145.2	Technology-based learning	Assignment, Quiz
17	quality functions	Evaluate community structures using quality functions.	DS 4145.2	Lecture-based teaching-learning	Assignment, Quiz
18	The Kernighan-Lin algorithm,	Understand the basics of community detection algorithms.	DS 4145.2	Lecture-based teaching-learning	Assignment, Quiz
19	Agglomerative algorithms1	Implement agglomerative for community detection.	DS 4145.2	Technology-based learning	Assignment, Quiz
20	Agglomerative algorithms2	Implement agglomerative for community detection.	DS 4145.2	Technology-based learning	Assignment, Quiz
Mid Term Examination					
21	Introduction to social influence,	Understand the concept of social influence in networks.	DS 4145.3	Lecture-based teaching-learning	Assignment, Quiz
22	Influence related statistics,	Analyze the role of social similarity in influence dynamics.	DS 4145.3	Lecture-based teaching-learning	Assignment, Quiz
23	social similarity	Evaluate influence measures and their applications.	DS 4145.3	Lecture-based teaching-learning	Assignment, Quiz
24	social influence,	Evaluate influence measures and their applications.	DS 4145.3	Lecture-based teaching-learning	Assignment, Quiz
25	Homophily	Analyze the role of homophily in influence dynamics.	DS 4145.3	Technology-based learning	Assignment, Quiz
26	Homophily2	Analyze the role of homophily in influence dynamics.	DS 4145.3	Technology-based learning	Assignment, Quiz
27	Existential Test for social influence,	Evaluate influence measures and their applications.	DS 4145.3	Lecture-based teaching-learning	Assignment, Quiz

28	Influence and actions,	Evaluate influence measures and their applications.	DS 4145.3	Technology-based learning	Assignment, Quiz
29	Influence and interactions,	Evaluate influence measures and their applications.	DS 4145.3	Technology-based learning	Assignment, Quiz
30	influence maximization in viral marketing.	Apply influence maximization techniques to optimize viral marketing strategies.	DS 4145.4	Lecture-based teaching-learning	Assignment, Quiz
31	Case Study -1	Apply theoretical knowledge to a practical case study. Use network analysis tools to explore and visualize network data.	DS 4145.4	Technology-based learning	Assignment, Quiz
32	Case Study -2	Apply theoretical knowledge to a practical case study. Use network analysis tools to explore and visualize network data.	DS 4145.4	Technology-based learning	Assignment, Quiz
33	Case Study -3	Apply theoretical knowledge to a practical case study. Use network analysis tools to explore and visualize network data.	DS 4145.4	Technology-based learning	Assignment, Quiz
34	Case Study - 4	Apply theoretical knowledge to a practical case study. Use network analysis tools to explore and visualize network data.	DS 4145.4	Technology-based learning	Assignment, Quiz
35	Syllabus Revision & Pre-Exam Doubts	Integrate knowledge across all outcomes	All COs	Group-teaching and learning	Mock Exam, Discussion
36	Syllabus Revision & Pre-Exam Doubts2	Integrate knowledge across all outcomes	All COs	Group-teaching and learning	Mock Exam, Discussion
End Term Examination					

H. Course Articulation Matrix:

CO	Statement	Correlation with Program Outcomes and Program Specific Outcomes														
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
DS4145.1	Presents interactive social applications on the web, and the types of analysis that are currently conducted in the study of social media,	2	1	1	1	2	1	1	1					1	1	
DS4145.2	Analyse the basics of network structures for beginners, including measuring methods for describing nodes, edges, and parts of the network,	1	1	1	1	1	1	1	1					1	1	
DS4145.3	Evaluate the major categories of social media applications or phenomena and shows how the techniques presented can be applied to analyse and understand the underlying data	2	2	2	2	1	1		1		1	1		1	1	1
DS4145.4	Understand information visualization, particularly network visualization techniques, and methods for using them to identify interesting features in a network, generate hypotheses for analysis, and recognize patterns of behaviour					1	1	1	1	1		1	1	1	1	1

1: Low Correlation; 2: Moderate Correlation; 3: Substantial Correlation

Course Coordinator

Name: Dr. Aprna Tripathi

MUJ ID: MUJ1400

Head of the Department

Prof. Akhilesh Kumar Sharma

Student Representative

Name:

Registration No: